6 A is the point with coordinates (5, 9)
B is the point with coordinates (d, 15)

The gradient of the line AB is 3

Work out the value of d.

$$\text{Gradient} = \frac{15 - 9}{d - 5} = 3$$

1 mark
Use change in y ÷ change in x

$$6 = 3(d - 5)$$

$\div 3$ $\quad$ $\div 3$

$$2 = d - 5$$

$+5$ $\quad$ $+5$

$$7 = d$$

1 mark
Rearrange to isolate d

7

1 mark
d = 7

(Total for Question 6 is 3 marks)